Linear Algebra II, Sheet 4, HT26

The Spectral Theorem and Applications

1. Show that every real symmetric matrix has a real symmetric cube root.

2. For $i = 1, 2$, find an orthogonal matrix P such that $P^T A_i P$ is diagonal where

$$A_1 = \begin{pmatrix} 3 & 2 \\ 2 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Hence sketch the conic

$$3x^2 + 4xy = 1,$$

carefully labelling the eigenvectors of A_1.

3. Let

$$X = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \\ 1 & 2 & 7 \end{pmatrix}$$

and let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ denote the columns of X in order.

(i) Apply the GSOP to $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ to produce an orthonormal basis $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$.

(ii) Hence write $X = PU$ where P is an orthogonal matrix and U is upper triangular.

(iii) Show that there are 8 upper triangular orthogonal 3×3 matrices.

(iv) Deduce that there are 8 ways to write $X = PU$ as in (ii).

4. (i) Say that $M = A^T A$ where A is a square matrix. Show that M is symmetric and that the eigenvalues of M are non-negative. [Hint: consider $|A\mathbf{v}|^2$ where $\mathbf{v}$ is an eigenvector of M.]

(ii) Say that N is a symmetric matrix with non-negative eigenvalues. Show that there exists A such that $N = A^T A$.

(iii) Find such A when

$$N = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

5. (i) Let A be a symmetric $n \times n$ matrix with maximum eigenvalue $\lambda_{\max}$ and minimum eigenvalue $\lambda_{\min}$. By making an orthogonal change of variable $\mathbf{x} = P\mathbf{y}$, show that

$$\lambda_{\min} |\mathbf{x}|^2 \leqslant \mathbf{x}^T A \mathbf{x} \leqslant \lambda_{\max} |\mathbf{x}|^2$$

for all $\mathbf{x} \in \mathbb{R}^n_{\text{col}}$.

(ii) Let

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}.$$

Sketch the quadric $\mathbf{x}^T A \mathbf{x} = 1$. What is the minimum value of $|\mathbf{x}|$ on this surface and where is it attained?

Starter

S1: Let A be an $n \times n$ matrix. Show that $A\mathbf{x} \cdot \mathbf{y} = \mathbf{x} \cdot A\mathbf{y}$ for all $\mathbf{x}$, $\mathbf{y} \in \mathbb{R}^n_{\text{col}}$ if and only if A is symmetric.

S2: Show that the volume of the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ is $4\pi abc/3$.

S3: Let A be an $m \times n$ matrix and $\mathbf{v} \in \mathbb{R}^n_{\text{col}}$. Show that $A^T A\mathbf{v} = \mathbf{0}$ if and only if $A\mathbf{v} = \mathbf{0}$. Deduce that the rank of $A^T A$ equals the rank of A.

Pudding

P1: Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \end{pmatrix}.$$

Find a 3×3 orthogonal matrix P and a 2×2 orthogonal matrix Q such that

$$PAQ = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \\ 0 & 0 \end{pmatrix},$$

where $\lambda_1 \geqslant \lambda_2 > 0$.

P2: With notation as in P1, the *pseudoinverse* of A is

$$A^+ = Q \begin{pmatrix} \lambda_1^{-1} & 0 & 0 \\ 0 & \lambda_2^{-1} & 0 \end{pmatrix} P.$$

(i) Determine A^+ where A is the matrix in P1.

(ii) Verify that $A^+ = \left(A^T A\right)^{-1} A^T$.

(iii) Let $\mathbf{b} = (1, 1, 1)^T \in \mathbb{R}^3_{\text{col}}$. Show that the equation $A\mathbf{x} = \mathbf{b}$ is inconsistent. Verify that $\mathbf{x}_0 = A^+\mathbf{b}$ satisfies

$$|A\mathbf{x} - \mathbf{b}| \geqslant |A\mathbf{x}_0 - \mathbf{b}| \qquad \text{for all } \mathbf{x} \in \mathbb{R}^2_{\text{col}}.$$

P3: (i) Let $\mathbb{R}[x]$ denote the space of polynomials with real coefficients. Show that

$$\langle f, g \rangle = \int_{-1}^{1} f(x)g(x)\,\mathrm{d}x$$

is an inner product on $\mathbb{R}[x]$.

(ii) Show that the linear map $L\colon \mathbb{R}[x] \to \mathbb{R}[x]$ given by

$$L(f) = \frac{\mathrm{d}}{\mathrm{d}x}\left[\left(1 - x^2\right)\frac{\mathrm{d}f}{\mathrm{d}x}\right]$$

satisfies $\langle Lf, g \rangle = \langle f, Lg \rangle$.

(iii) Show that there is a sequence of polynomials $P_0, P_1, P_2, \ldots$ such that

- $\deg P_n = n$.
- The P_n are orthonormal with respect to the inner product given in (i).
- The P_n are eigenvectors of L.